

Storing data with DVC and OneDrive

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Usage notes for data storage within university constraints.

Background

The University of Iceland has an “unlimited” OneDrive setup for staff / graduate students. This is generally not very helpful, but in conjunction with `rclone` [^fn:1] and `dvc` can be made to be of more use.

Baseline

Like most of my newer projects, `pixi` is used to manage the tooling required, along with `git-subrepo` for handling local changes to projects.

```
pixi add dvc rclone
```

As before, we will assume `gnome-keyring` is present, and then since `rclone` supports secrets stored in password managers we have[^fn:2]:

```
secret-tool store --label="dvc" service "rclone" project "gprd_dimer"
```

Which can then simply be set to `RCLONE_PASSWORD_COMMAND` and used directly:

```
export RCLONE_PASSWORD_COMMAND='secret-tool lookup service "rclone" project "gprd_dimer"'
rclone config --ask-password=false
```

We don’t want duplication. So there has to be a question of if we want to use the artifact uploaded directly to an S3 instance or if we want to use local and let DVC handle putting it on a bucket if need be.

Rather than use the local NAS approach of letting DVC take an absolute path to an `rclone mount` point, it is a bit nicer to use an `rclone server` WebDAV instance instead.

```
pixi add dvc-webdav
# In one terminal
rclone serve webdav HI0neDrive:/.dvcstore --vfs-cache-mode full --addr unix:///tmp/dvc.hi.socket
# In another
dvc remote add -d HI0neDrive webdavs:///tmp/dvc.hi.socket
```

Where we assume that `rclone` is configured as per the [OneDrive instructions](#). Since there are no sensitive details in any of the above commands, it is alright to have DVC store this configuration and track it with `git`.

Handling SSL errors

Annoyingly, it turns out that even with `ssl_verify false`, an SSL record layer error will be thrown if the WebDAV instance doesn’t offer SSL.

```
dvc remote modify hionedrive ssl_verify false
dvc push
Collecting
|0.00 [00:00,    ?entry/s]
Pushing
ERROR: unexpected error - [SSL] record layer failure (_ssl.c:1000): [SSL] record layer failure (_ssl.c:1000)
```

Which means that we need to setup a key pair, with the defaults, which is OK since it will only be used locally:

```
# Keep hitting enter to accept the defaults
openssl ecparam -out ec_key.pem -name prime256v1 -genkey
openssl req -new -key ec_key.pem -x509 -nodes -days 365 -out cert.pem
```

This can now be used to serve the WebDAV instance:

```
rclone serve webdav HIOneDrive://.dvcstore --vfs-cache-mode full --addr localhost:9677 --cert cert.pem --key ec_key.pem
```

DVC doesn't care about the exact key-pair, and with the current setup browsers will (correctly) raise `net::ERR_CERT_AUTHORITY_INVALID` warnings [^fn:3] but will now merrily accept the connection (since `ssl_verify` is still off):

```
dvc push
Collecting
Pushing
196 files pushed
|0.00 [00:00,    ?entry/s]
```

Other `dvc` idiosyncracies

- Ignore folders in the `.gitignore`, else `dvc` may complain that the relevant `.dvc` is ignored by `git`
 - i.e. to ignore `best_trial_8/` don't ignore `best_trial*` but use `best_trial*/`

Conclusions

This isn't necessarily the most reliable setup as written, given that the University reclaims access pretty aggressively, however when coupled with the [Materials Archive](#) and a [strong compression technique](#) for personal storage via `borg` it is fairly resilient.